1. Identity

Document	STAR-PKG-STAR-TDB-EN · v2.1 · 2026-11-01
Package	star-tdb
Version	0.1.1
Capability	star.time.tdb
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_tdb

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_tdb-0.1.1-py3-none-any.whl
```

FILE	SHA-256
star_tdb-0.1.1-py3-none-any.whl	8f7477569a4c6b5f8a57da74b98c9f95bded2cfe60497e678647d78029674471
star_tdb.py	126fafbabc0d9fd6e288042c8abbb282516c5ac9859485d09a12688d4d642e35
star_tdb-0.1.1-source.zip	b37f47039c52170e4e6731a984e465f61ab1b731e4f4c43122c0bcabe62f28c3

5. What it does, and its limits

TDB - TT for a TT Julian date, geocentre. Standard library only.

Topocentric terms (observer position), sub-microsecond accuracy, dates outside 1600-2200.

- `tdb_minus_tt(jd_tt)` → `seconds`, USNO Circular 179 (Kaplan 2005) eq. 2.6, 7 terms.
- Accuracy within 10 microseconds of the full Fairhead & Bretagnon series over 1600-2200.
- Outside 1600-2200 raises `ValueError` (no extrapolation beyond stated validity).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_tdb_0_1_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-tdb}: TDB - TT with the USNO Circular 179 series, verified against ERFA full series within 10 microseconds},
  version = {0.1.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_tdb/},
```

```
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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Related: space activities and export control · acceptable use · evidence and limits · licence texts